Advanced Machine Learning, ETH Zurich

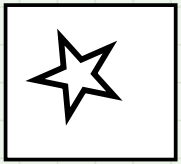
Convolutional neural networks

Carlos Cotrini

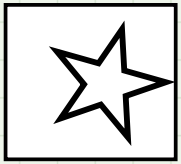
Autumn semester 2026

Entry 1

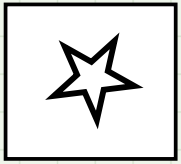
Stars and planets



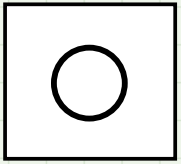
Star



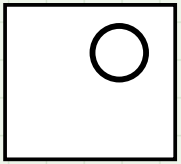
Star



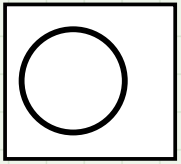
Star



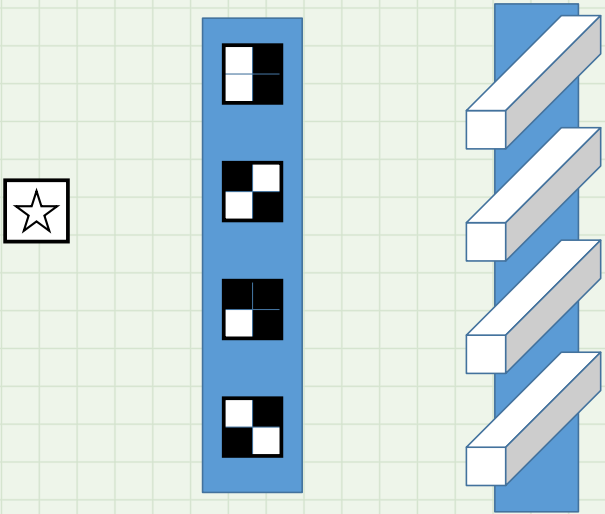
Planet



Planet

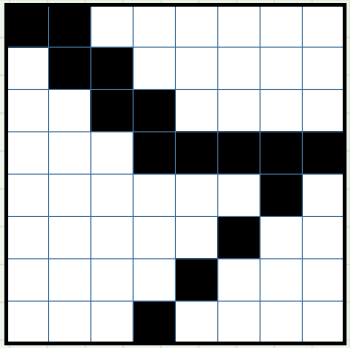


Planet



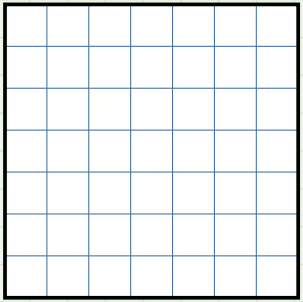
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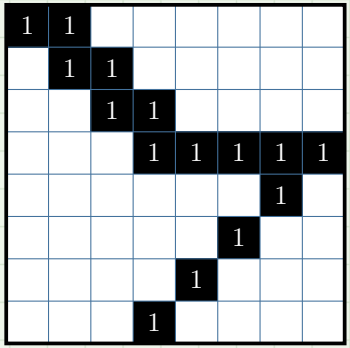
Convolutional filters





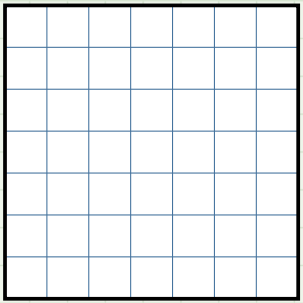
Convolutional filter

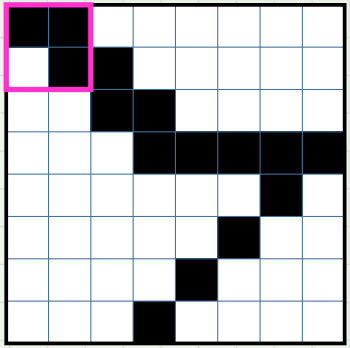






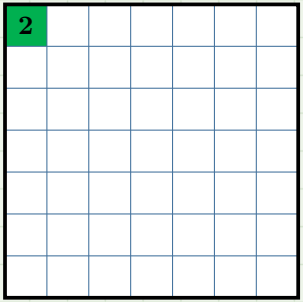
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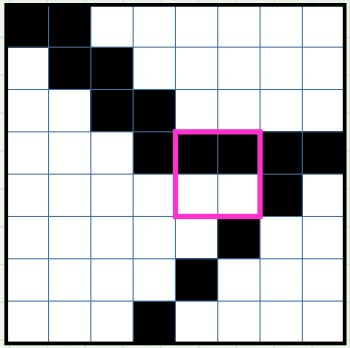






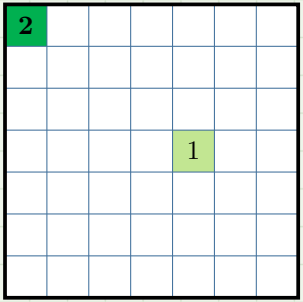
Convolutional filter

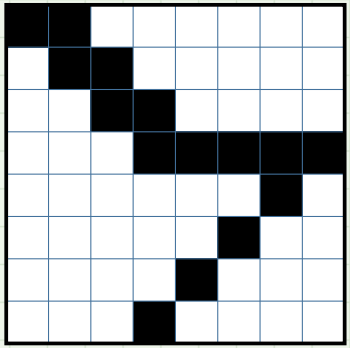






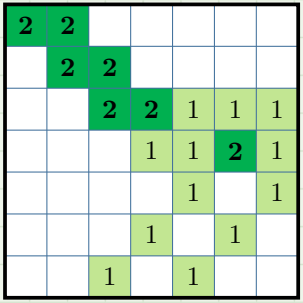
Convolutional filter

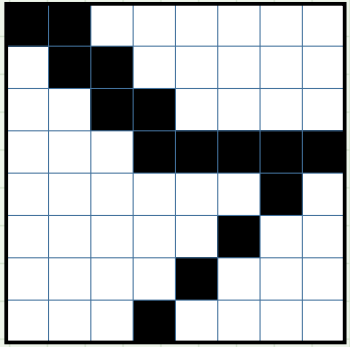






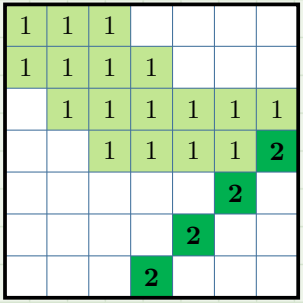
Convolutional filter







Convolutional filter



1	3	5
1	2	4
6	3	2

Input

1	2
−5	0

ReLU

Convolutional filter

Output

1	3	5
1	2	4
6	3	2

Input

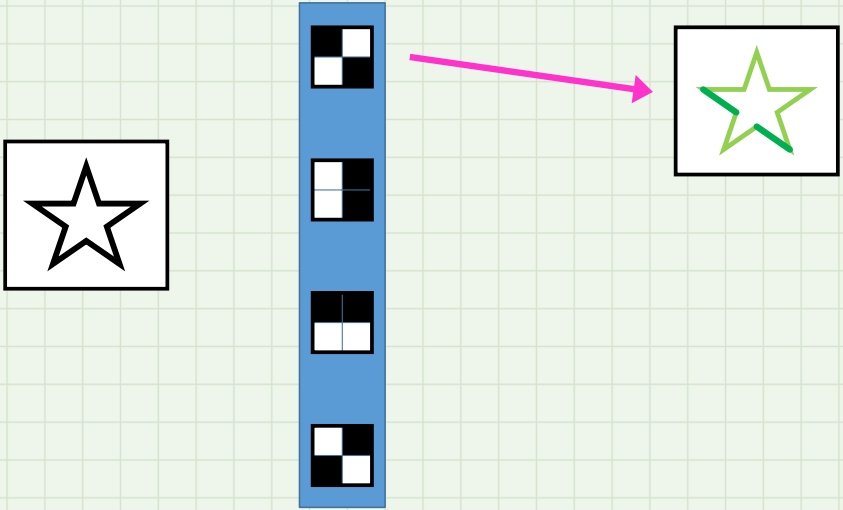
1	2
−5	0

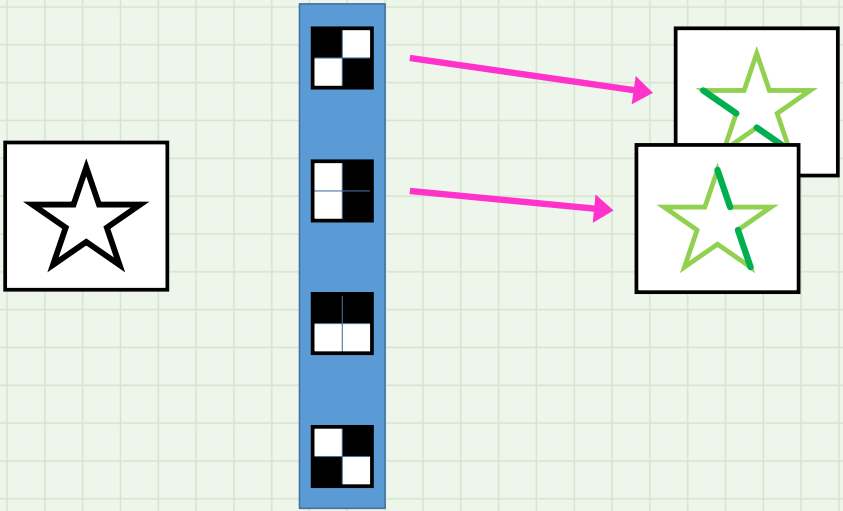
ReLU

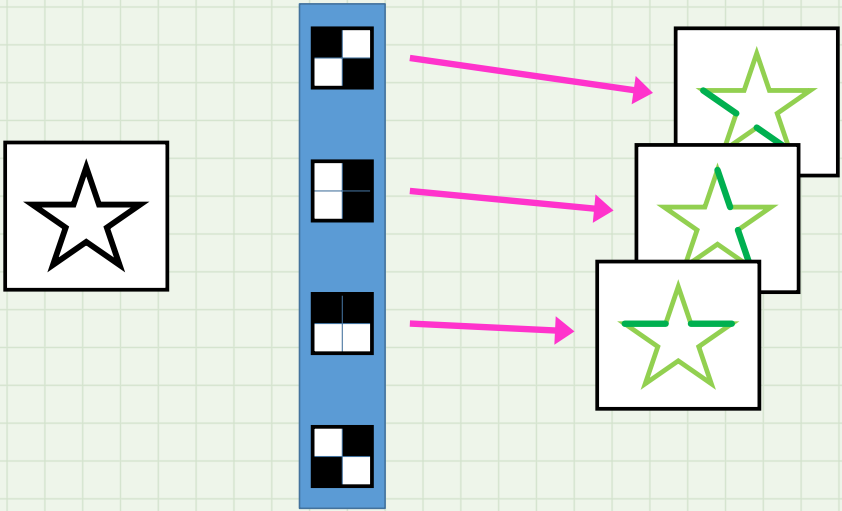
Convolutional filter

2	3
0	0

Output

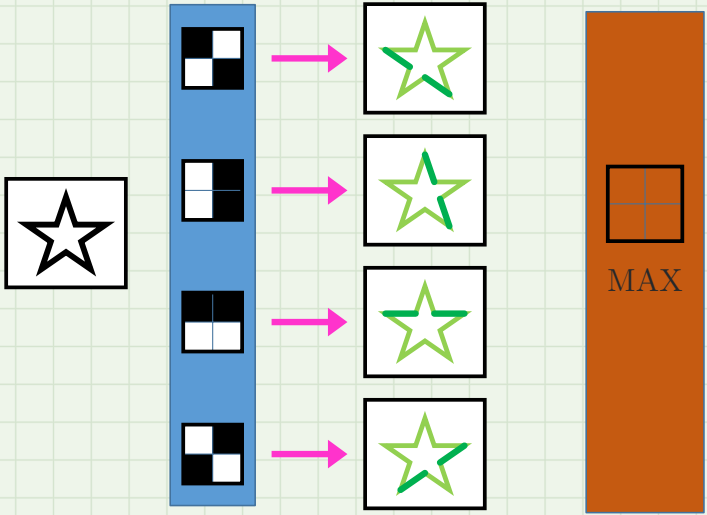


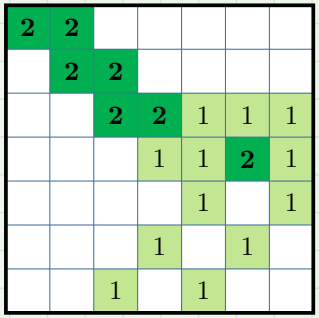




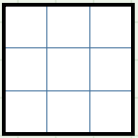
Entry 3

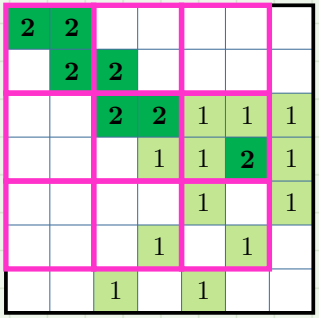
Pooling

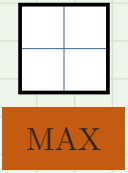

MAX

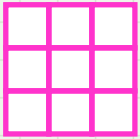

2 2
2 2
2 2 1 1 1
1 1 2 1
1 1
1 1
1 1

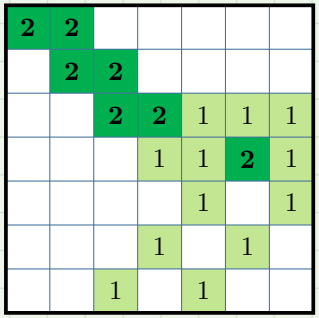

MAX



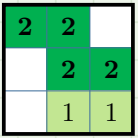

2 2
2 2
2 2 1 1 1
1 1 2 1
1 1
1 1
1 1

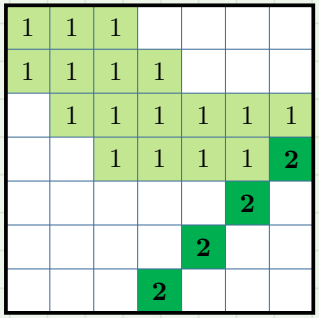

MAX

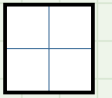



2 2
2 2
2 2 1 1 1
1 1 2 1
1 1
1 1
1 1

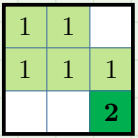

MAX


2 2
2 2
1 1



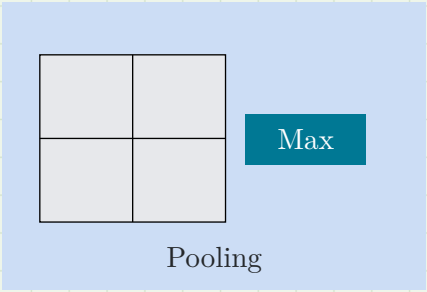


MAX



1	0	5	3	1
4	0	0	2	2
6	1	0	1	0
0	0	0	2	4
3	0	1	2	1

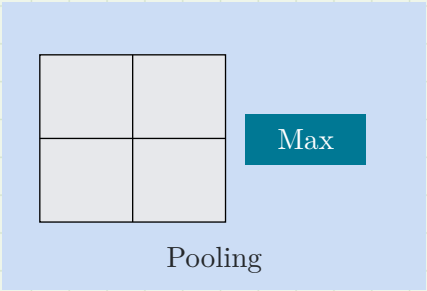
Input



Output

1	0	5	3	1
4	0	0	2	2
6	1	0	1	0
0	0	0	2	4
3	0	1	2	1

Input

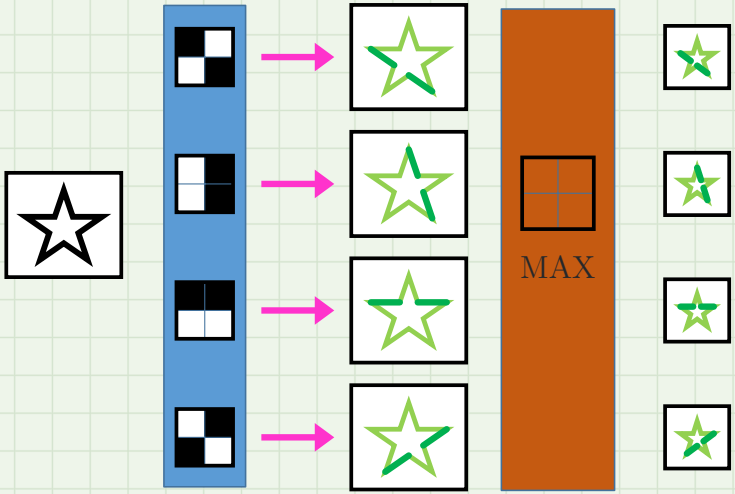


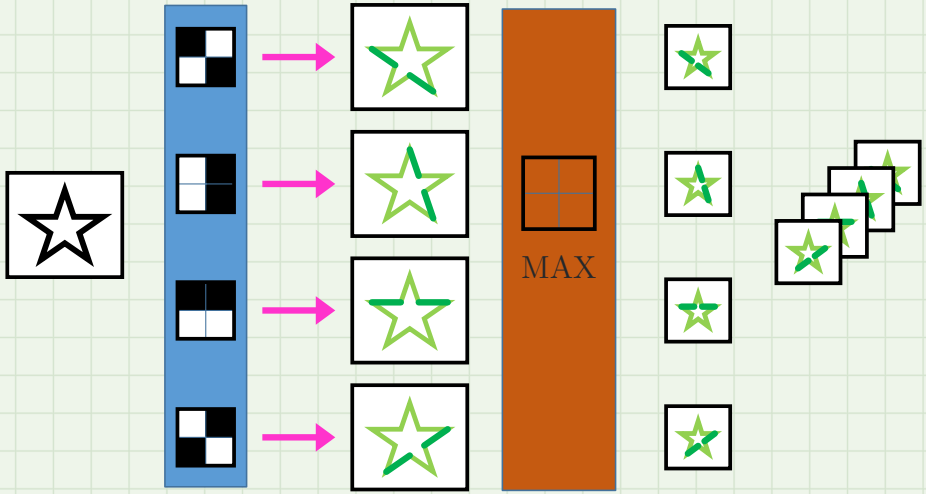
4	5
6	2

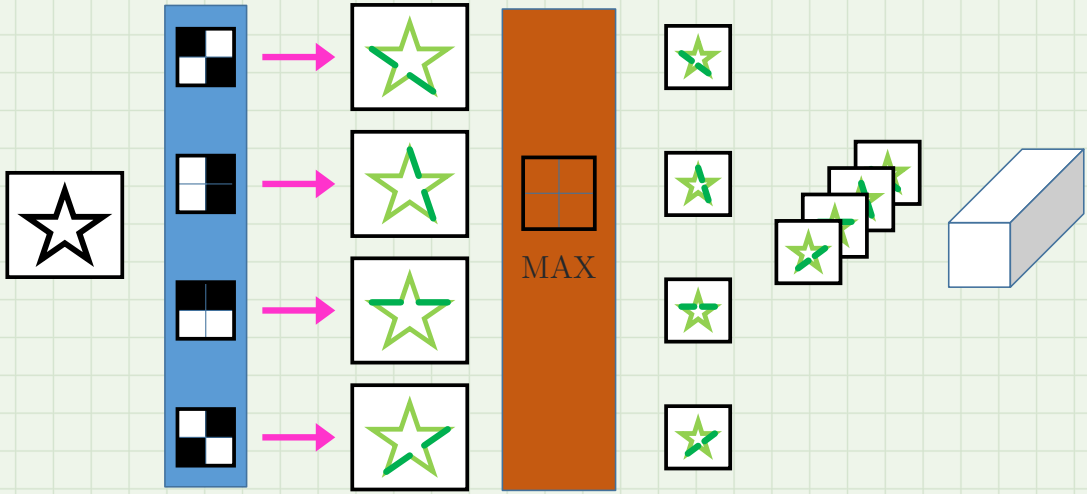
Output

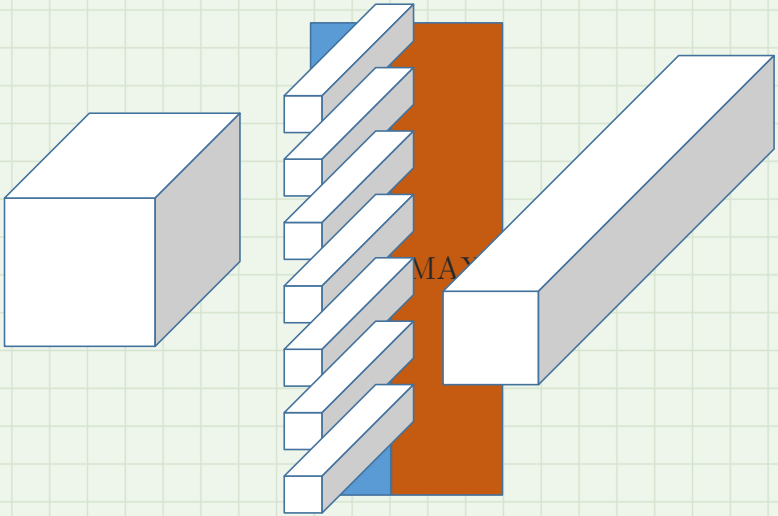
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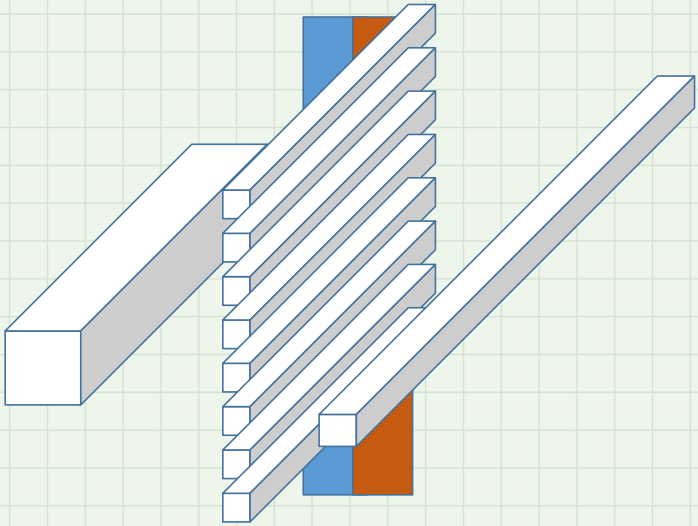
The network

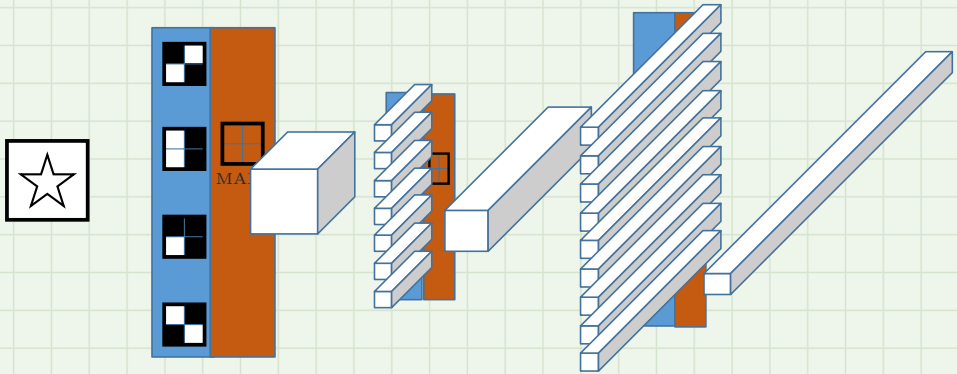

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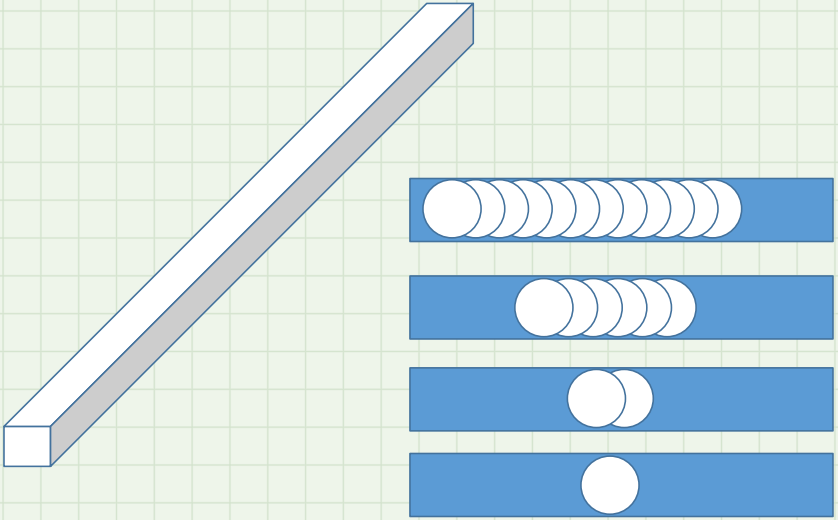

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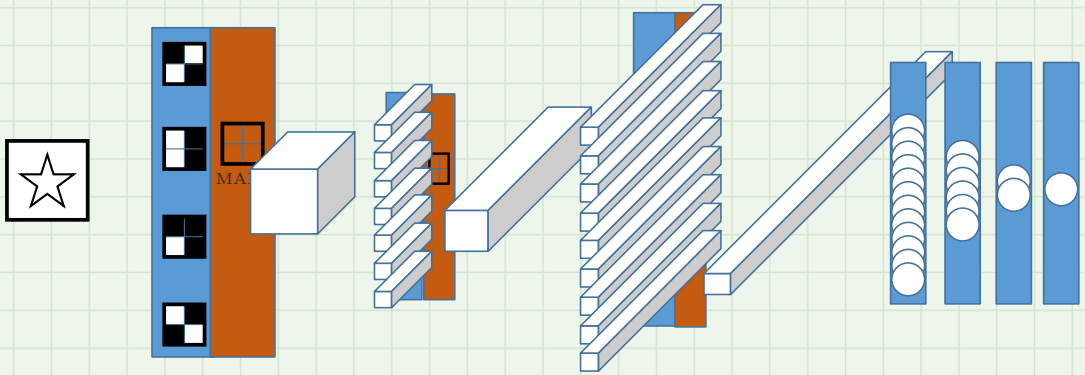

MAX


MAX



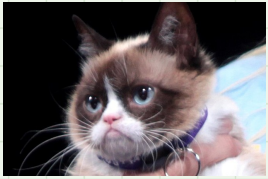




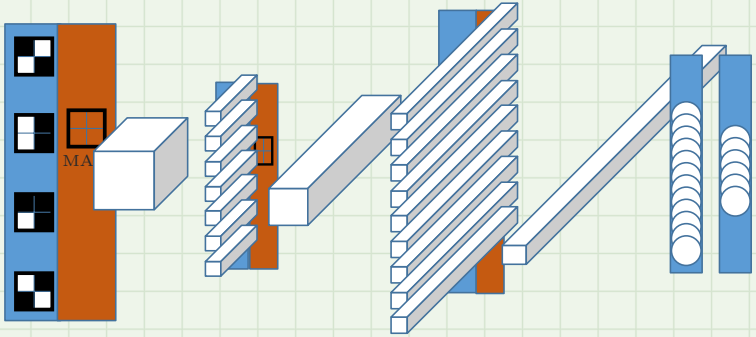


Entry 5

Image recognition and softmax

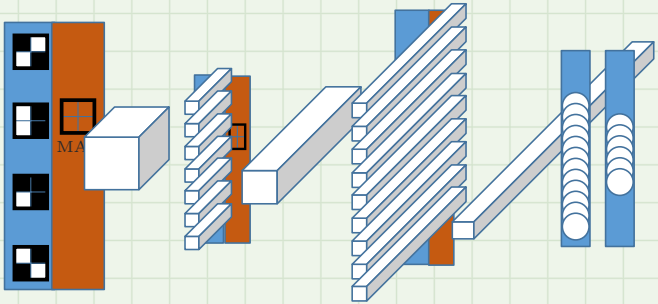


Grumpy Cat: Gage Skidmore, CC BY-SA 3.0, Wikimedia Commons.





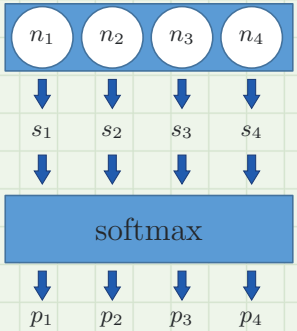
Rabbit: JM Ligero Loarte, CC BY 3.0. Cat: Alvesgaspar, CC BY-SA 3.0. Dog: Dog breed facts, CC BY-SA 4.0. All from Wikimedia Commons.





Rabbit: JM Ligero Loarte, CC BY 3.0. Cat: Alvesgaspar, CC BY-SA 3.0. Dog: Dog breed facts, CC BY-SA 4.0. All from Wikimedia Commons.

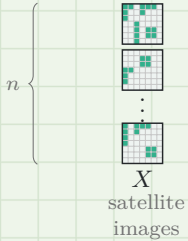
- In binary classification tasks, we want to recognize if a particular given object belongs to one of two classes. For these tasks, the CNN has one neuron at the end, whose activation function is the sigmoid function. It produces a value between 0 and 1, indicating the probability that the input belongs to one particular class.
- In multi-class classification tasks, we want to recognize if a particular given object belongs to one of K classes. For these tasks, the CNN has K neurons $n_1, n_2, \ldots, n_K$ at the end, each of them producing a score of how much each neuron believes that the object belongs to one of these classes. After these K neurons, we have a softmax function that transforms these values into probabilities $p_1, p_2, \ldots, p_K$. Each probability p_i indicates the probability that the object belongs to class i.



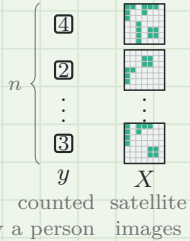
Entry 6

Training a CNN to count islands

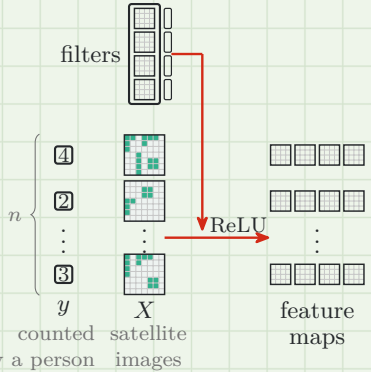
A person counts the islands in many satellite images. The network learns to give the same counts: it predicts them, measures its errors and corrects its filters.



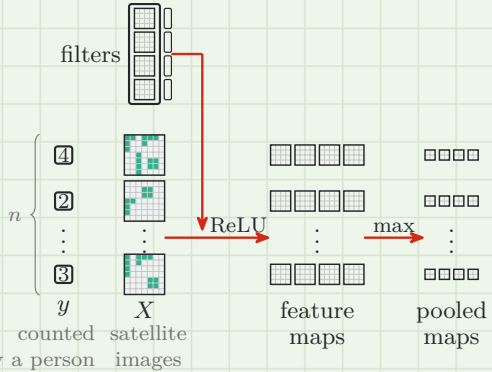
A person counts the islands in many satellite images. The network learns to give the same counts: it predicts them, measures its errors and corrects its filters.



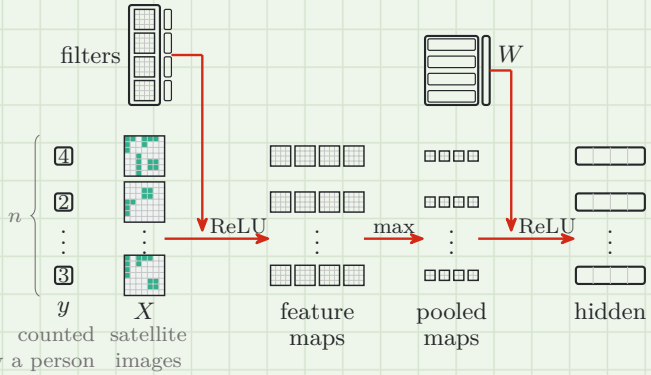
A person counts the islands in many satellite images. The network learns to give the same counts: it predicts them, measures its errors and corrects its filters.



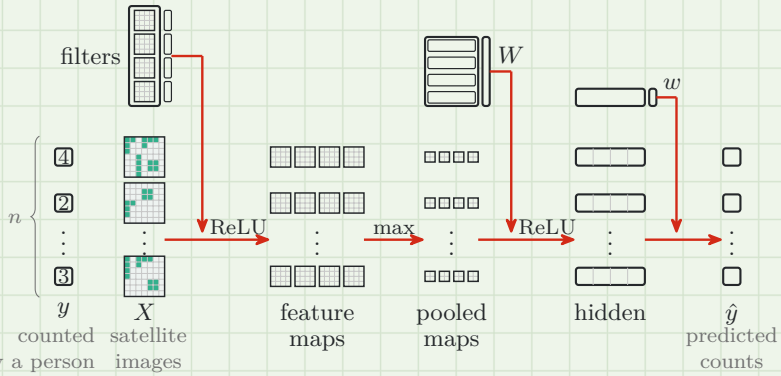
A person counts the islands in many satellite images. The network learns to give the same counts: it predicts them, measures its errors and corrects its filters.



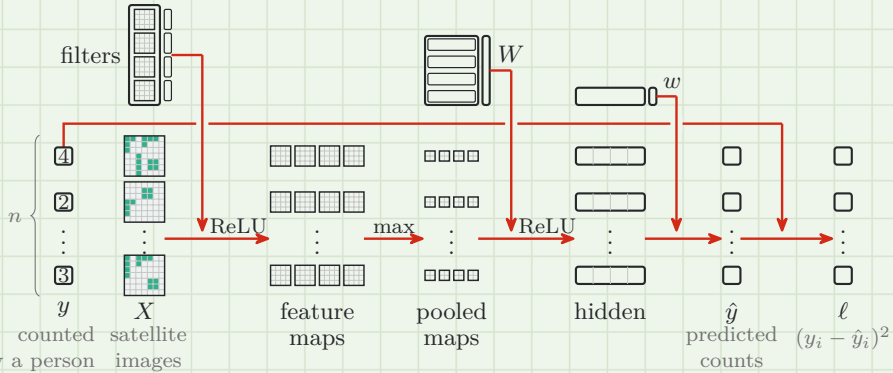
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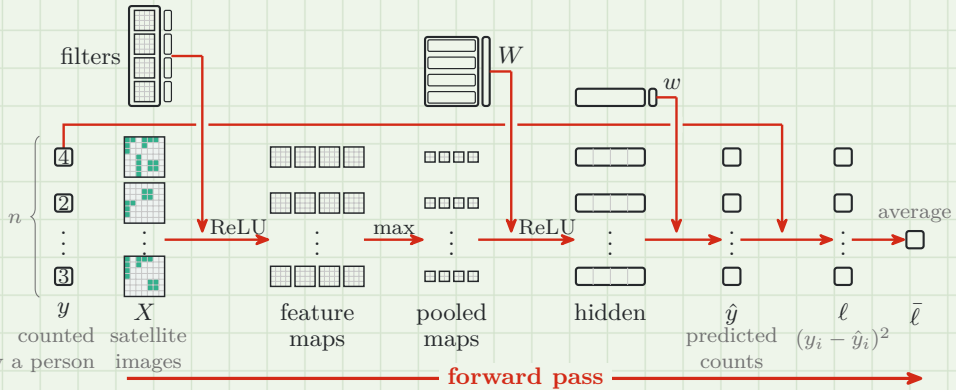
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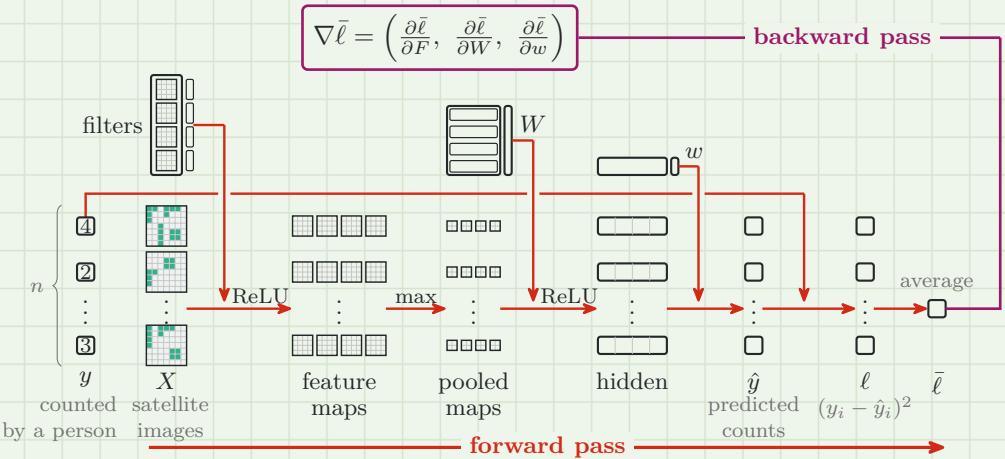
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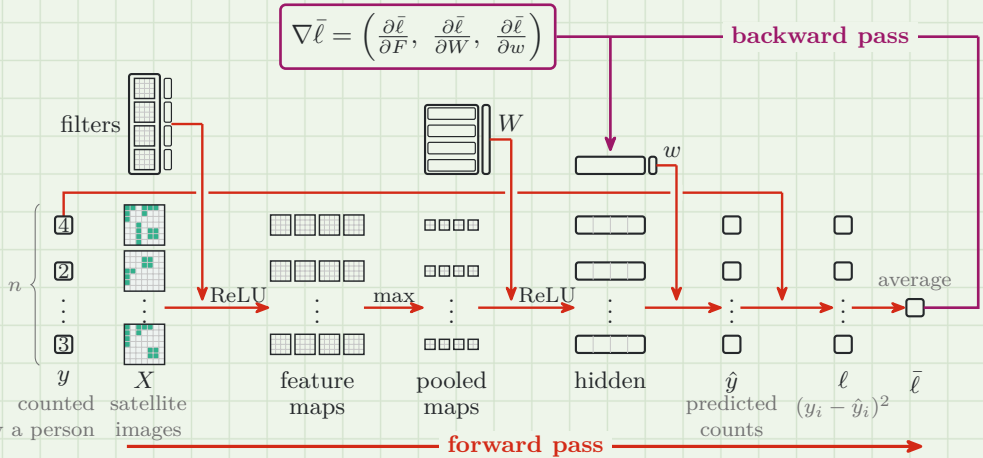
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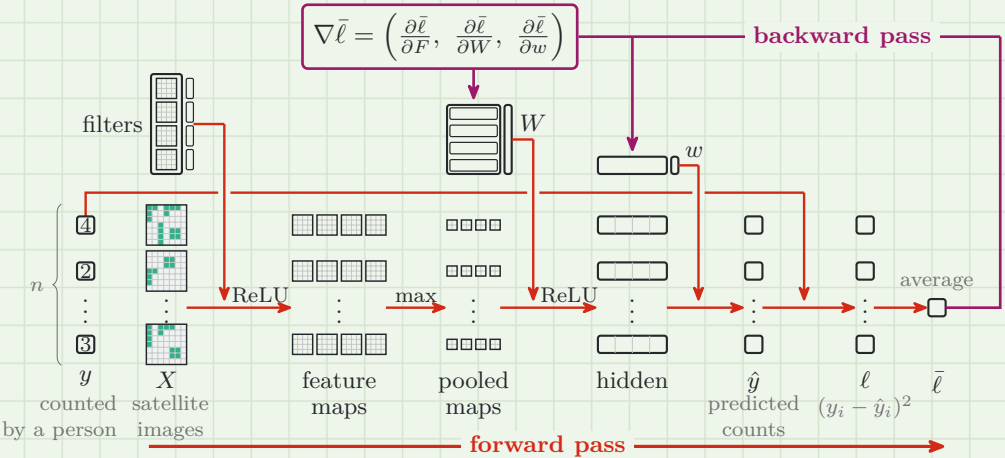
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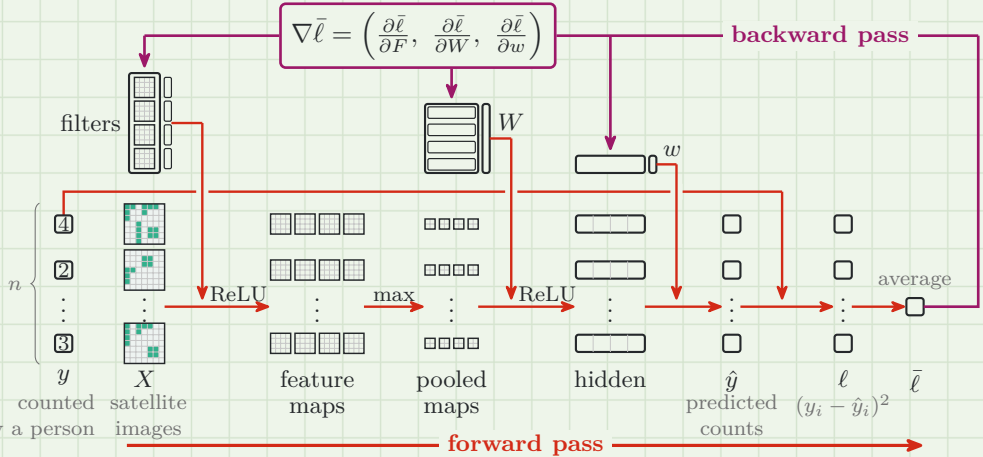
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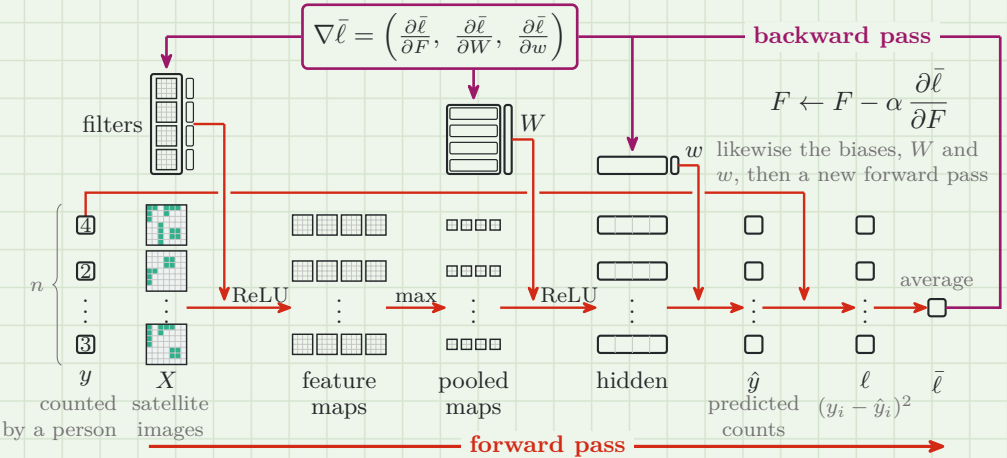
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When training starts, nobody knows what the filters should look like. Every filter is a grid of 4 by 4 unknown numbers, and each filter has one more unknown, its bias.

filter 1

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

filter 2

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

filter 3

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

filter 4

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

4 filters of 4×4: 64 unknown numbers

When training starts, nobody knows what the filters should look like. Every filter is a grid of 4 by 4 unknown numbers, and each filter has one more unknown, its bias.

filter 1

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

?

bias

filter 2

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

?

bias

filter 3

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

?

bias

filter 4

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

?

bias

4 filters of 4×4: 64 unknown numbers
and 4 biases: 68 numbers to guess in the filters

When training starts, nobody knows what the filters should look like. Every filter is a grid of 4 by 4 unknown numbers, and each filter has one more unknown, its bias.

filter 1

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

? bias

filter 2

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

? bias

filter 3

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

? bias

filter 4

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

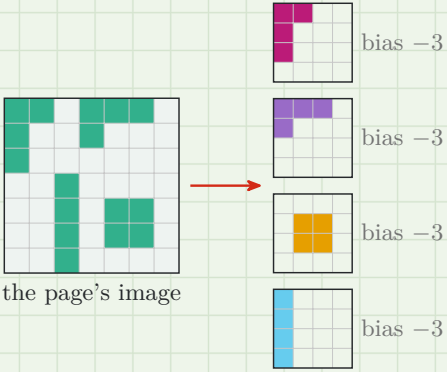
? bias

4 filters of 4×4: 64 unknown numbers

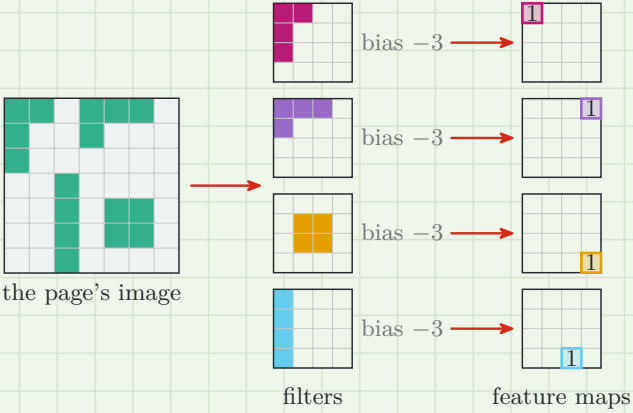
and 4 biases: 68 numbers to guess in the filters

Each update $F \leftarrow F - \alpha\, \partial\bar{\ell}/\partial F$ moves every one of them a little.

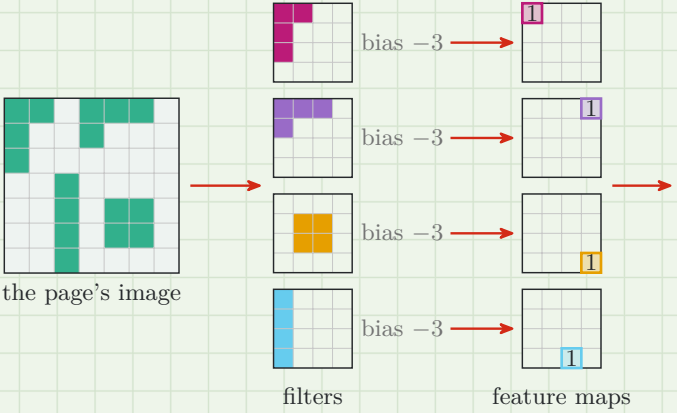
One good choice is known: the four Tetris tiles, each with bias −3. Each tile finds its one island in the page's image, so adding up the maps gives the count.



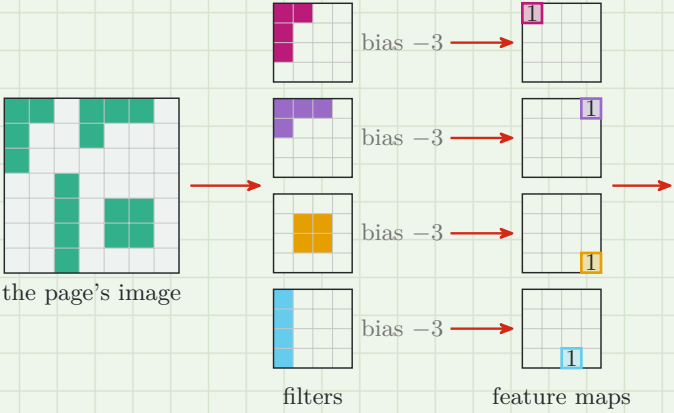
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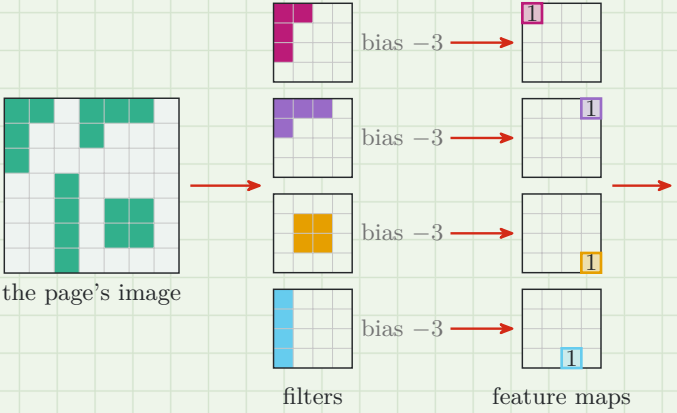
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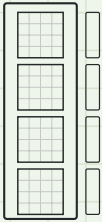
a person also counts $y = 4$

$$\hat{y} = 1 + 1 + 1 + 1 = 4$$

max pooling keeps each 1, and the dense layers add them up

Training starts from blank filters and finds numbers that count as well, from the counted images alone.

Training guesses every number in a block of weights: the entries of the filters, the biases and the weights of the two dense layers. Max pooling has nothing to guess.



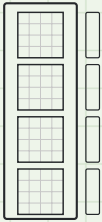
filters

$4 \times 4 \times 4 = 64$

$+ 4$ biases

68

Training guesses every number in a block of weights: the entries of the filters, the biases and the weights of the two dense layers. Max pooling has nothing to guess.

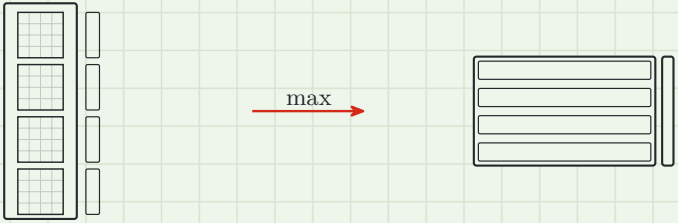


filters
$4 \times 4 \times 4 = 64$
$+$ 4 biases
68

max

max pooling
nothing
to guess
0

Training guesses every number in a block of weights: the entries of the filters, the biases and the weights of the two dense layers. Max pooling has nothing to guess.

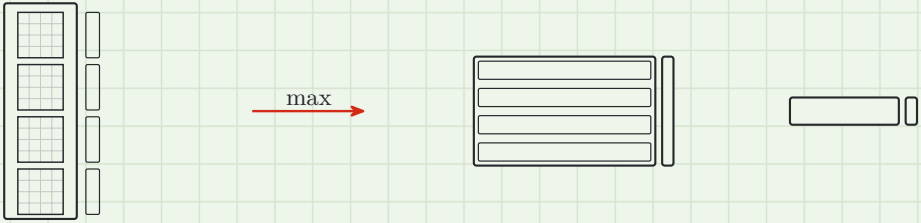


filters
$4 \times 4 \times 4 = 64$
$+ 4$ biases
68

max pooling
nothing
to guess
0

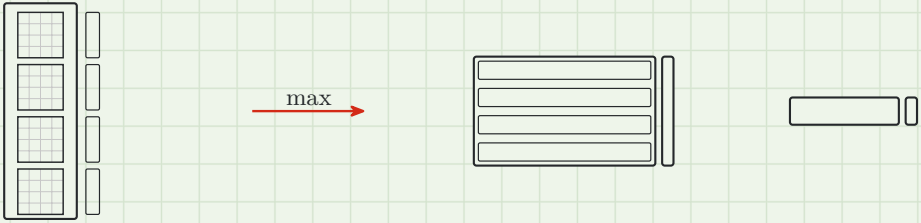
dense layer W
$16 \times 4 = 64$
$+ 4$ biases
68

Training guesses every number in a block of weights: the entries of the filters, the biases and the weights of the two dense layers. Max pooling has nothing to guess.



filters	max pooling	dense layer W	output w
$4 \times 4 \times 4 = 64$	nothing	$16 \times 4 = 64$	$4 \times 1 = 4$
$+ 4$ biases	to guess	$+ 4$ biases	$+ 1$ bias
68	0	68	5

Training guesses every number in a block of weights: the entries of the filters, the biases and the weights of the two dense layers. Max pooling has nothing to guess.



filters	max pooling	dense layer W	output w
$4 \times 4 \times 4 = 64$	nothing	$16 \times 4 = 64$	$4 \times 1 = 4$
$+ 4$ biases	to guess	$+ 4$ biases	$+ 1$ bias
68	0	68	5

$$68 + 0 + 68 + 5 = 141 \text{ numbers to guess}$$